

SAI VINEETH AKULA

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SUMMARY

Data Engineer / Data Analyst with **6+ years of experience** designing scalable **batch and streaming data pipelines** across **healthcare, IoT, and CRM** platforms. Expertise in **Apache Spark, Airflow, Kafka, Python, and SQL**, with experience building **cloud-native data platforms on AWS and Azure**. Proven ability to process **multi-terabyte datasets**, develop **ML-ready feature pipelines**, and optimize distributed workflows supporting **analytics, machine learning, and BI systems**. Experienced in **data lake architectures, real-time ingestion pipelines, and enterprise data quality frameworks**.

PROFESSIONAL EXPERIENCE

Blue Cross Blue Shield (BCBS) | USA

Jul 2024 – Present

Data Engineer / Data Analyst

- Engineered **Apache Spark ETL normalization pipelines** resolving schema drift across **18TB+ healthcare claims and eligibility datasets** in **Azure Data Lake**, reducing pipeline failures **35%** and stabilizing ingestion for analytics reporting.
- Instituted **SQL reconciliation frameworks** within **Apache Airflow DAG workflows**, standardizing eligibility records across **50M+ members** and improving healthcare data quality **30%**.
- Directed **provider attribution pipelines** using **Airflow and Azure Synapse Analytics**, improving reliability of healthcare analytics workflows **40%**.
- Established **data quality monitoring frameworks** with schema validation, anomaly detection, and completeness checks supporting **Power BI healthcare dashboards**.
- Curated **feature engineering datasets** for predictive healthcare analytics, reducing ML preprocessing effort **25%**.
- Facilitated collaboration with **data science and actuarial teams**, delivering **ML-ready datasets** and improving model data freshness **30%**.
- Provisioned **Docker-based ingestion services** with **GitHub Actions CI/CD**, reducing pipeline release cycles **40%** across **Azure environments**.
- Delivered **Spark SQL and Python ETL pipelines** generating analytics-ready datasets for **Power BI dashboards** supporting claims and provider analytics.

HCL Technologies | India

Nov 2021 – Aug 2022

Data Engineer

- Spearheaded **NLP sentiment analytics pipelines** using **TensorFlow, Python, and Apache Spark**, processing **5M+ CRM interactions** for enterprise customer analytics.
- Formulated **text feature engineering pipelines** (tokenization, TF-IDF, embeddings) improving sentiment classification accuracy **20%**.
- Integrated **Salesforce CDC events** with **Azure Service Bus**, enabling **real-time ingestion** into **MSSQL analytics tables**.
- Enhanced **Apache Kafka streaming pipelines** by optimizing consumer parallelism and partitioning, increasing throughput **40%**.
- Reengineered **SQL Server schemas and indexing strategies**, improving analytical query performance **35%**.
- Segmented **Azure Synapse Analytics tables** using temporal partitioning to accelerate **Power BI and Tableau** reporting queries.
- Streamlined **ETL pipelines** converting **Salesforce CRM datasets** into curated analytics tables supporting retention and customer behavior analysis.

Amazon | India

Aug 2018 – Jul 2021

Data Engineer

- Constructed **Python and Apache Spark ingestion pipelines** processing **10M+ IoT telemetry events daily** into an **Amazon S3 data lake** for operational analytics.
- Generated **feature engineering pipelines** producing telemetry aggregates and anomaly indicators improving forecasting models **18%**.
- Coordinated **Spark preprocessing pipelines** through **Apache Airflow orchestration** for reliable distributed data processing.
- Enforced **SQL validation checks and reconciliation rules** to resolve telemetry inconsistencies impacting ML datasets.
- Refined **Amazon Redshift schemas** using optimized distribution styles and sort keys, improving analytical query performance **45%**.
- Safeguarded **Spark pipelines** with monitoring, retry logic, and completeness checks ensuring reliable analytics datasets.
- Containerized **Spark workloads using Docker** and deployed through **Jenkins CI/CD pipelines**.
- Produced **analytics-ready telemetry datasets** powering operational dashboards and enterprise BI reporting.

Robosoft Technologies | India

Aug 2017 – Aug 2018

Data Analyst

- Consolidated fragmented operational logs into normalized relational datasets using **SQL ETL workflows**, transforming **3M+ monthly records** for BI reporting.
- Performed **data profiling, statistical validation, and anomaly detection**, improving reliability of analytics datasets **25%**.
- Structured **relational data models** supporting reporting workloads and improving dashboard query efficiency.
- Standardized reusable **SQL transformation pipelines** for data cleansing and feature generation supporting analytics initiatives.
- Cataloged **data dictionaries and schema documentation**, strengthening enterprise data governance.
- Compiled curated analytics datasets via **SQL ETL workflows**, enabling operational dashboards and reducing manual reporting **30%**.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL (PostgreSQL, MySQL), Java
- Data Processing & Frameworks:** Apache Spark, Spark SQL, Spark Structured Streaming, Batch Processing, Streaming Processing
- Machine Learning & AI:** Feature Engineering, Data Preparation for ML, Model Inference Pipelines, NLP Pipelines, TensorFlow, ML Data

Validation, Model Support Pipelines

- **Orchestration & Workflow Management:** Apache Airflow, DAG Design, Job Scheduling, Dependency Management
- **Data Warehousing & Storage:** Amazon Redshift, Snowflake, BigQuery, Amazon S3, Azure Data Lake Storage, Azure Synapse Analytics
- **Databases:** PostgreSQL, MySQL, Microsoft SQL Server, MongoDB
- **Streaming & Messaging Systems:** Apache Kafka, Azure Service Bus
- **Cloud Platforms:** AWS, Azure, GCP
- **Containers, DevOps & CI/CD:** Docker, Kubernetes exposure, Jenkins, GitHub Actions, CI/CD Pipelines
- **Data Modeling & Quality:** Schema Design, Partitioning Strategies, Data Validation, Reconciliation, Data Quality Monitoring
- **AI-Assisted Automation:** AI-assisted debugging, Automated data validation, Workflow automation for analytics and ML pipelines
- **ETL / ELT & Reporting:** ETL Pipelines, ELT Pipelines, Power BI, Tableau, SQL-Based Reporting, Operational Dashboards

PROJECTS

Real-Time Environmental Data Pipeline for Data Center Energy Monitoring:

- Built **Kafka + Spark Streaming pipeline** processing **8–10M energy telemetry events/day** into a centralized data lake.
- Orchestrated batch aggregations using **Apache Airflow** and SQL transformations into a cloud warehouse.
- Enabled **12-month analytics across 20+ regional facilities** for energy monitoring insights.

Smart Healthcare IoT Data Integration Platform :

- Implemented a scalable **ETL architecture** to ingest streaming **IoT sensor data (wearables, remote monitors)** using **Kafka** and **Spark Structured Streaming**, standardizing **10 M+ daily events** into a unified **schema** for **clinical analysis**.
- Established robust **data quality validation** and **schema evolution controls** with **Airflow orchestration** and **SQL reconciliation checks**, improving **data reliability** for **predictive health models** in **preventive care scenarios**.

EDUCATION

Master of Science in Computer Science The University of Texas at Arlington Texas, USA	<i>Aug 2022 - May 2024</i>
Bachelor of Science in Computer Science SR University Warangal, India	<i>Aug 2013 - May 2017</i>

CERTIFICATIONS

- **AWS Certified Cloud Practitioner**
- **Tableau for Data Science**